

Dataset Details

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Field	Details
Dataset name	InfraDriftGuard Synthetic Cloud Configuration Drift Dataset (v1.0)
Source	Generated by this project (src/backend/dataset_generator.py) from 20 expert-defined drift scenarios modelled on AWS security best practices and CIS AWS Foundations Benchmark controls.
URL	GitHub repository: dataset/processed/dataset.csv and dataset/raw/raw_records.json (InfraDriftGuard_Cloud_Project_2026)
Size	dataset.csv ≈ 68 KB; raw_records.json ≈ 1.3 MB
Number of records	711 records (Low 180, Medium 176, High 175, Critical 180)
Number of features	9 input features + 1 label (plus 3 metadata columns: record_id, scenario_id, group_id)
Data type	Categorical (resource_type, changed_attribute, security_sensitivity, change_magnitude, port_exposure), Boolean (public_exposure, encryption_change, privilege_change), Integer (drift_frequency); label: ordinal categorical (Low < Medium < High < Critical). Raw file: nested JSON desired/actual states.
License	MIT License (same as the repository); contains no personal or customer data.
Purpose of using the dataset	Train and evaluate the drift risk classifier and SHAP explanations. Public datasets of labelled configuration drift do not exist (real drift data is private to each organisation), so a controlled synthetic dataset with documented ground-truth rules is used, and the live AWS pipeline is validated separately.

Features

Feature	Type	Meaning
resource_type	Categorical	s3_bucket / security_group / iam_policy
changed_attribute	Categorical	Which attribute(s) drifted, e.g. block_public_access
security_sensitivity	Ordinal	Intrinsic sensitivity of the attribute (Low/Medium/High)
public_exposure	Boolean	Did the change expose the resource to the internet?
encryption_change	Boolean	Was encryption weakened or removed?
privilege_change	Boolean	Were permissions escalated or a control removed?
port_exposure	Ordinal	none / internal / 0.0.0.0/0
change_magnitude	Ordinal	None/Low/Medium/High size of the change
drift_frequency	Integer	How often this resource has drifted before (from DynamoDB history)
risk_label (target)	Ordinal	Low / Medium / High / Critical

Drift scenarios

ID	Resource	Drift	Risk
S1	S3	Public access block disabled	Critical
S2	S3	Bucket ACL changed to public-read	Critical
S3id	S3	Server-side encryption disabled	High
S4	S3	Bucket policy principal widened to *	Critical
S5	S3	Versioning disabled	Medium
S6	S3	Access logging disabled	Medium
S7	S3	Lifecycle/retention rule removed	Low
E1	Security group	SSH (22) opened to 0.0.0.0/0	Critical
E2	Security group	RDP (3389) opened to 0.0.0.0/0	Critical
E3	Security group	New port opened publicly	High
E4	Security group	Outbound widened to allow-all	High
E5	Security group	No change (control)	Low
E6	Security group	Internal port widened to a range	Medium
E7	Security group	Cross-SG reference added	Medium
I1	IAM	Policy escalated to Action:* Resource:*	Critical
I2	IAM	Cross-account trust added	Critical
I3	IAM	MFA condition removed	High
I4	IAM	Explicit Deny flipped to Allow	Critical
I5	IAM	Group membership escalated to admin	Critical
I6	IAM	Permissions boundary removed	High

Preprocessing required

- Read with keep_default_na=False so the literal category “None” (no-drift control) is not turned into a missing value.
- Drop metadata columns (record_id, scenario_id, group_id) to prevent the model memorising scenarios.
- Ordinal-encode security_sensitivity, change_magnitude and port_exposure in severity order.
- One-hot encode resource_type and changed_attribute (unknown categories ignored at inference).
- Cast Booleans to 0/1; pass drift_frequency through unchanged.
- Encode the label with a fixed severity order (Low=0 ... Critical=3).
- Two splits: stratified random 80/20 (568/143) and scenario-group-aware (592/119) to measure leakage.